

SESSION OBJECTIVES

THE STUDENT PHYSICIANS WILL BE ABLE TO:

1. **Demonstrate** the components of the **history and physical** examination of the wrist and hand
2. **Evaluate** the patient utilizing **inspection, palpation, active range of motion, passive range of motion, strength (motor), neurologic, vascular, and osteopathic examination** of the wrist and hand
3. **Apply** testing and signs of the wrist and hand including **Phalen's test, Tinel's sign, thumb abduction, Allen test, and the Finklestein test**
4. **Distinguish normal versus abnormal examination** findings.
5. **Anticipate** and discuss common abnormal examination findings including **carpal tunnel syndrome, scaphoid fracture, and De Quervain's Tenosynovitis, and Arthritis**.
6. Recall the appropriate way to **document** common normal and abnormal findings of history and examination intake by **writing a SOAP note within 9 minutes**.

What are we talking about today?

- History considerations for the MSK complaint (and wrist/ hand)
- Physical exam of the wrist and hand
- Testing for abnormalities of wrist and hand
- Osteopathic Treatments
- SOAP note writing

LIFE LESSON TIME!!!

- After asking your history questions...
 - Time to think about what your potential diagnosis list will look like
- How will this list look after you perform your physical exam?
 - Will things be ruled in or excluded?
 - What other data will you need from testing?

Wrist and Hand – EXAM!

- Inspection
- Palpation
- Active Range of Motion (AROM)
- Passive Range of Motion (PROM)
- Strength Testing
- Abnormality Confirmation Testing
- Neurovascular Exam
- Osteopathic Exam/ Treatment

Wrist and Hand - PALPATION

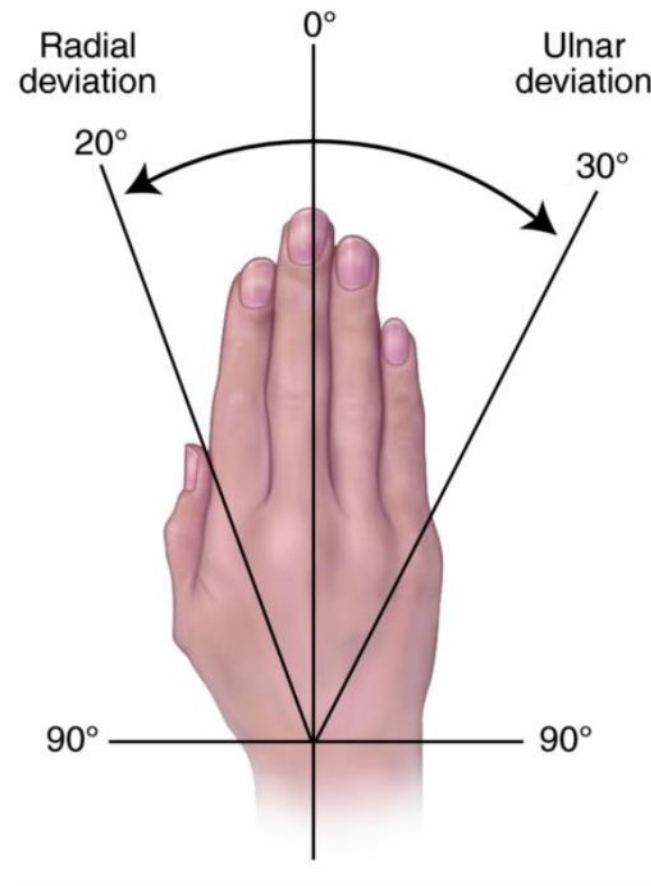
- Palpating the Wrist



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Wrist and Hand– ACTIVE Range of Motion

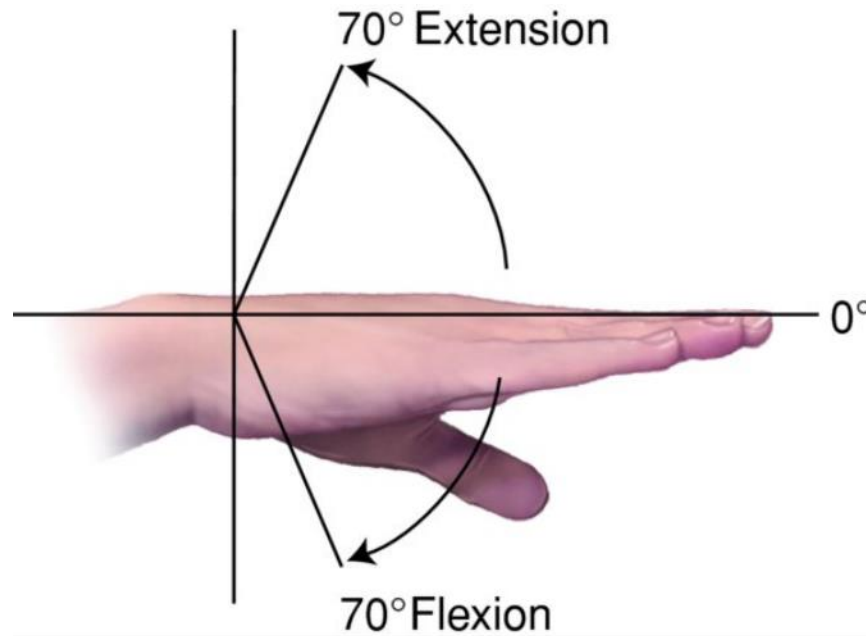
- Wrist Radial and Ulnar Deviation



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Wrist and Hand– ACTIVE Range of Motion

- Wrist Flexion and Extension



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Wrist and Hand – Strength Testing

- Graded on scale out of 5...
- 5/5 – full strength
- 4/5 – diminished strength
- 3/5 – movement against gravity
- 2/5 - movement without gravity
- 1/5 - involuntary movements
- 0/5 - no movement at all!!!

Wrist and Hand – Grip Strength

- Tests intrinsic hand muscles and finger flexors...



- Decreased grip strength is a **positive test** for weakness of the finger flexors and/or intrinsic muscles of the hand.
- Grip weakness plus wrist pain are often present in *de Quervain tenosynovitis*.

- TINEL's – tapping over the median nerve
 - Reproduces pain and paresthesias in the distribution of the **median nerve**
- Tests for Carpel Tunnel Syndrome (CTS)

- **Finklestein's Test** – stressing the Extensor Pollicus Brevis (EPB) and Abductor Pollicus Longus (APL)
 - Fold the thumb into the hand and then adduct the hand with some force
- Test for **DeQuervain's Tenosynovitis**

- **Thumb Abduction Test** - extend the thumb. Have patient abduct the thumb against resistance.
 - If there is weakness present, it may indicate Carpal Tunnel Syndrome.
- Why???

- Check capillary refill in several fingers
 - Compress the nail bed of a digit for 2-3 seconds. Release pressure, and immediately observe the return of circulation (pink color)
- Normal time is less than 3 seconds
- Maybe compare to other side if unsure

- Allen Test – checking for adequate perfusion to the hand
 - Occlude both the radial and ulnar arteries. Have patient make a fist/open hand several times. Release occlusion over radial artery. Monitor hand color.
- **Repeat** for Ulnar Artery.

- **Physical exam:**
- Gen: Alert, Oriented to person, place, time; No apparent distress. Well groomed.
- MSK:
 - Inspection: No gross abnormalities appreciated
 - Tenderness: Located to midline wrist right hand
 - ROM: Non-affected digits show full passive and active range of motion
 - Tendons: Non-affected digits glide freely without evidence of lag or incompetence or triggering
 - Neurovascular: Symmetric pulses and sensation bilaterally; Decreased grip strength right hand
 - Provocation tests: Positive Tinel's, Positive Phalen's
- Osteopathic: Decreased motion to lunate, capitate of right hand

WHAT NOW?

Wrist and Hand – Documentation (Comprehensive)

Objective MSK

Observation: Pt. was point tender over carpal tunnel on R wrist w/o crepitation. There was a little bit of swelling in the wrist but not a whole lot. No obvious deformity was found, however, the R thenar eminence was smaller due to atrophy of the muscles.

<u>PROM/AROM:</u> AROM:	Wrist Flexion: 83	Finger flexion(MCP): 70
	Wrist Extension: 87	Finger extension(MCP):43
	Wrist radial deviation: 20	Finger abduction: 25
	Wrist ulnar deviation: 30	Finger adduction: 23
PROM:	Wrist Flexion: 85	Finger flexion(MCP): 75
	Wrist extension: 90	Finger extension(MCP): 47
	Wrist radial deviation: 27	Finger abduction: 28
	Wrist ulnar deviation: 33	Finger adduction: 25
<u>RROM/MMT:</u>	Wrist flexion: 5/5	Finger adduction: 5/5
	Wrist extension: 5/5	Thumb MP and IP flexion: 5/5
	Finger MP flexion: 5/5	Thumb MP and IP extension: 5/5
	Finger PIP and DIP flexion: 5/5	Thumb abduction: 5/5
	Finger MP extension: 5/5	Thumb adduction: 5/5
	Finger abduction: 5/5	

Neurological Testing (Reflexes, dermatomes, myotomes):

Dermatomes- C6, C7, C8 (All WNL)
Reflexes- Biceps (C5-C6) (WNL)
Brachioradialis (C6) (WNL)
Triceps (C7) (WNL)

Cardiovascular Testing:

Radial artery: 2/4
Ulnar artery: 2/4

Special Tests: Tinel's sign at wrist (+), Phalens (+),

Osteopathic: As above, Decreased motion to lunate, capitate of right hand